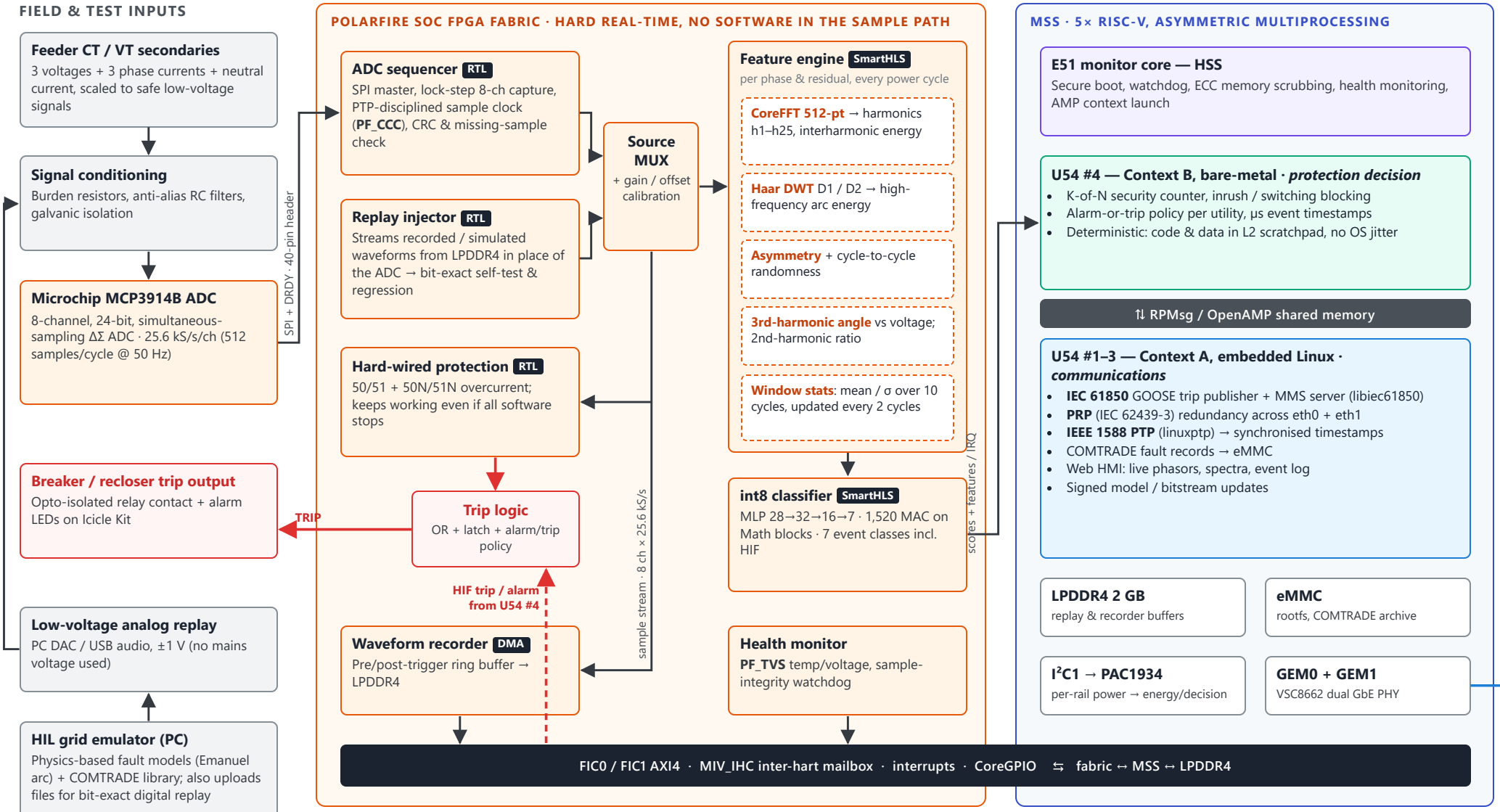
ArcSentinel — System Block Diagram

FPGA-accelerated detection & protection against high-impedance (downed / arcing conductor) faults

Track 2 · PolarFire® SoC Icicle Kit (MPFS250T)

Team ArcSentinel: Hemanth Chilakala (lead), Tanmaya Kumar · v1.1 · Oct 2026



LEGEND

Fabric (RTL / SmartHLS) · Bare-metal RISC-V · Linux RISC-V · Monitor core · External / test · Trip path

TOOLCHAIN & BUILD

Libero SoC + SmartHLS + SoftConsole · Icicle Kit reference design · Yocto/Buildroot AMP machine (Linux harts 1–3, bare-metal hart 4) · Python golden model

LATENCY BUDGET (TARGETS)

Instantaneous overcurrent: sample → trip output (fabric)	**< 1 ms**
Cycle features + int8 inference (fabric)	**< 50 µs**
Sampling jitter (clock-derived)	**< 10 ns**
HIF declared after onset (model: median 0.28 s)	**≤ 0.5 s**
Decision → GOOSE frame on both LANs	**< 4 ms**
Messages lost when one LAN fails (PRP)	**0**

STATION NETWORK

PRP LAN A + LAN B → substation SCADA / HMI and remote breakers & reclosers (GOOSE subscribers)

98.3%	**2.5%**	**0 / 720**
HIFs detected, 0.4–40 A (pre-proposal model)	51N + inrush restraint on same faults	false alarms on look-alike events

ArcSentinel — Data Path, Mixed-Criticality Partition & Timing

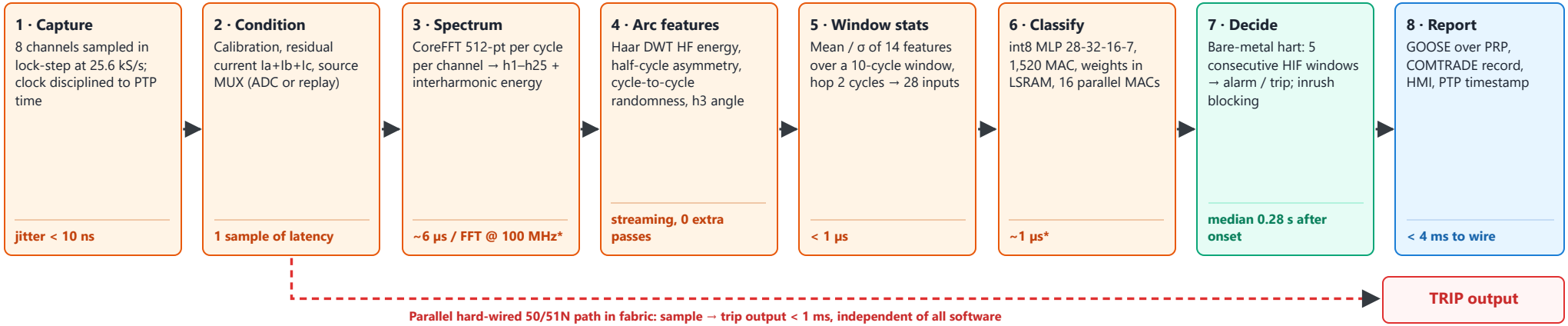
How each power cycle travels from the ADC to a trip decision, and why each function lives where it does

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A. Per-cycle fabric pipeline

every 20 ms per channel (50 Hz); everything runs in parallel, no CPU involvement



B. Three independent layers of protection

our core novelty vs prior HIF devices

Layer 1 — FPGA fabric (safety-critical, hard real-time)

Sampling, feature extraction, int8 inference and classic overcurrent protection run as dedicated logic. No shared caches, interrupts or scheduler, so latency is identical every cycle. If every processor halts, overcurrent protection still trips the breaker.

Layer 2 — Bare-metal U54 hart (deterministic decision)

Applies security logic to the fabric's per-window scores: persistence counters, inrush and switching blocking, alarm-versus-trip policy. Runs from L2 scratchpad with no OS, isolated from Linux by PMP and AMP context separation.

Layer 3 — Linux on 3 U54 harts (communications, non-critical)

IEC 61850 GOOSE/MMS, PRP redundancy on both GbE ports, PTP, COMTRADE, web HMI and signed updates. A Linux crash or network attack cannot block a trip; the E51 monitor restarts the context.

C. Hardware / software partition

Function	Where	Why there
ADC capture & timing	Fabric RTL	Lock-step sampling with zero jitter; no missed samples
FFT, wavelet, randomness features	Fabric (CoreFFT + SmartHLS)	Every channel analysed every cycle, in parallel, at fixed latency
int8 classifier	Fabric (SmartHLS)	Deterministic inference on Math blocks; same C++ runs on CPU for baseline
50/51/N overcurrent	Fabric RTL	Safety function that must not depend on software
Security & trip policy	Bare-metal U54 #4	Flexible logic with deterministic timing
GOOSE / MMS / PRP / PTP / HMI	Linux U54 #1–3	Rich protocol stacks; not in the critical path
Boot, watchdog, ECC scrub	E51 (HSS)	Supervises and recovers the other contexts

* Pre-synthesis estimates, to be confirmed by Libero timing reports in development week 4. Detection figures come from our pre-proposal Python golden model on **synthetic** data (2,520 events: 1,680 train / 840 held-out test, independent random seeds). Real-data validation (public Victorian VCIT field-test data) and hardware-in-the-loop replay are part of the development plan. The pipeline is written for 50 Hz and switches to 60 Hz with one parameter change (512 samples/cycle kept by retiming the sample clock).